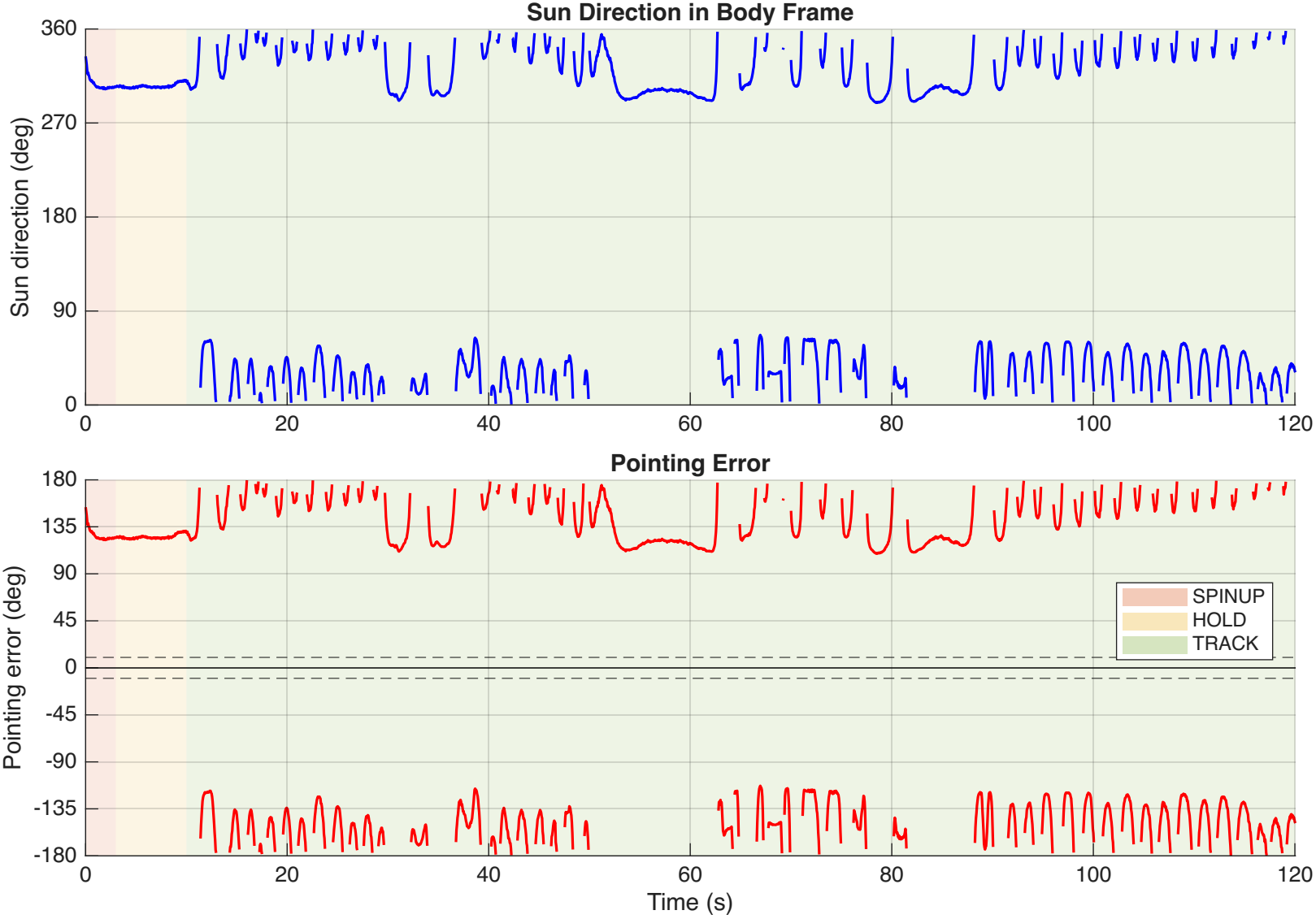
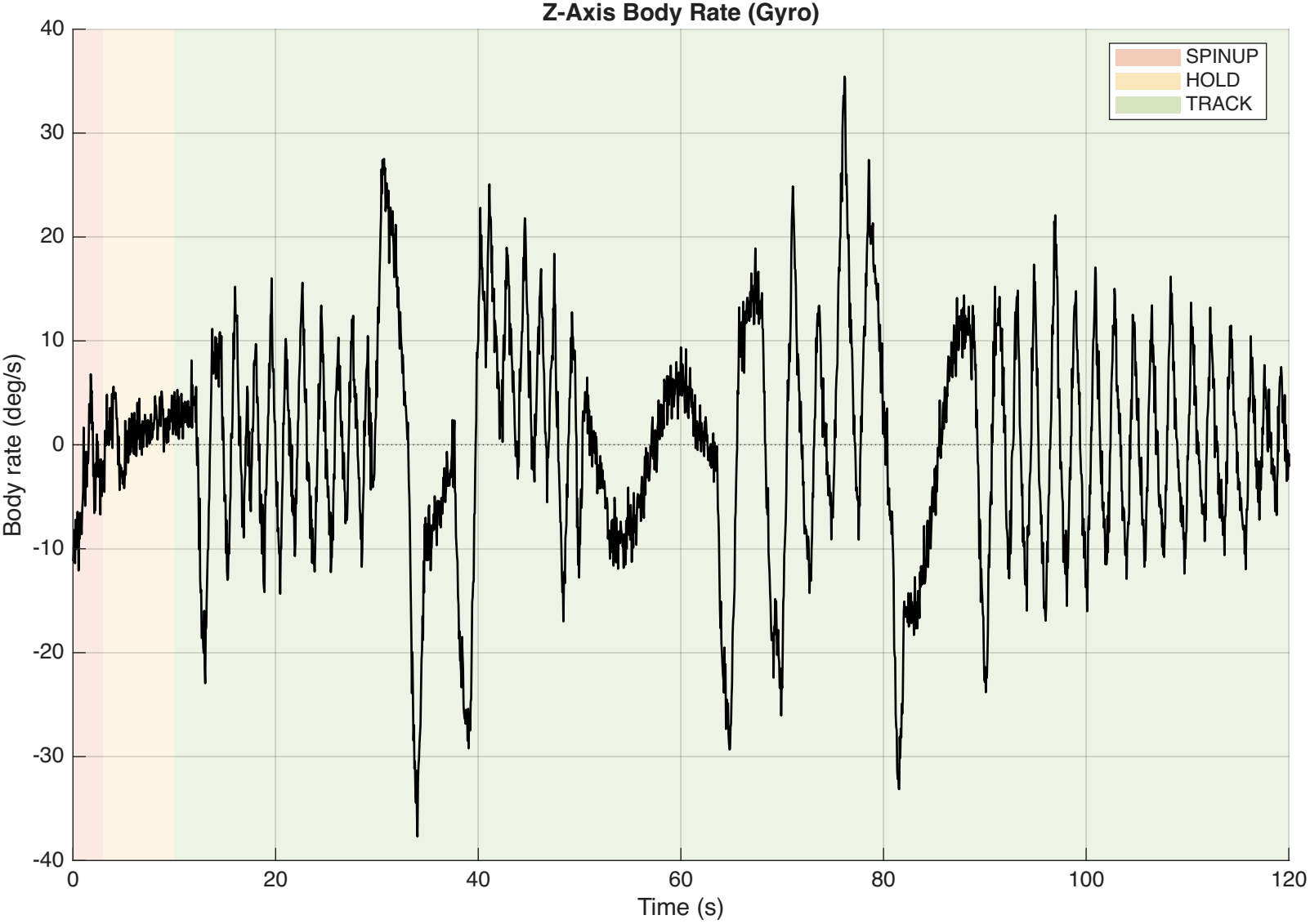
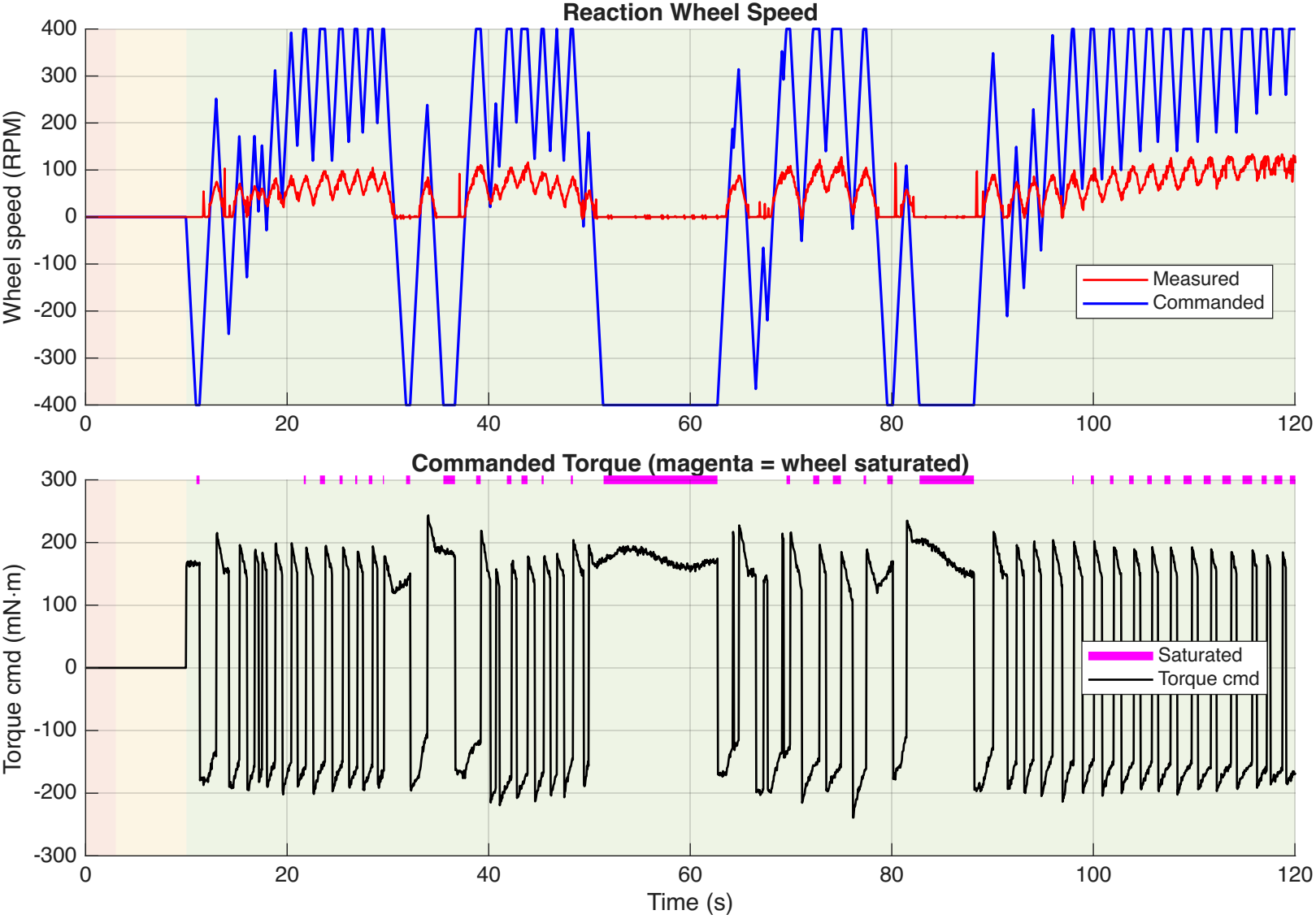
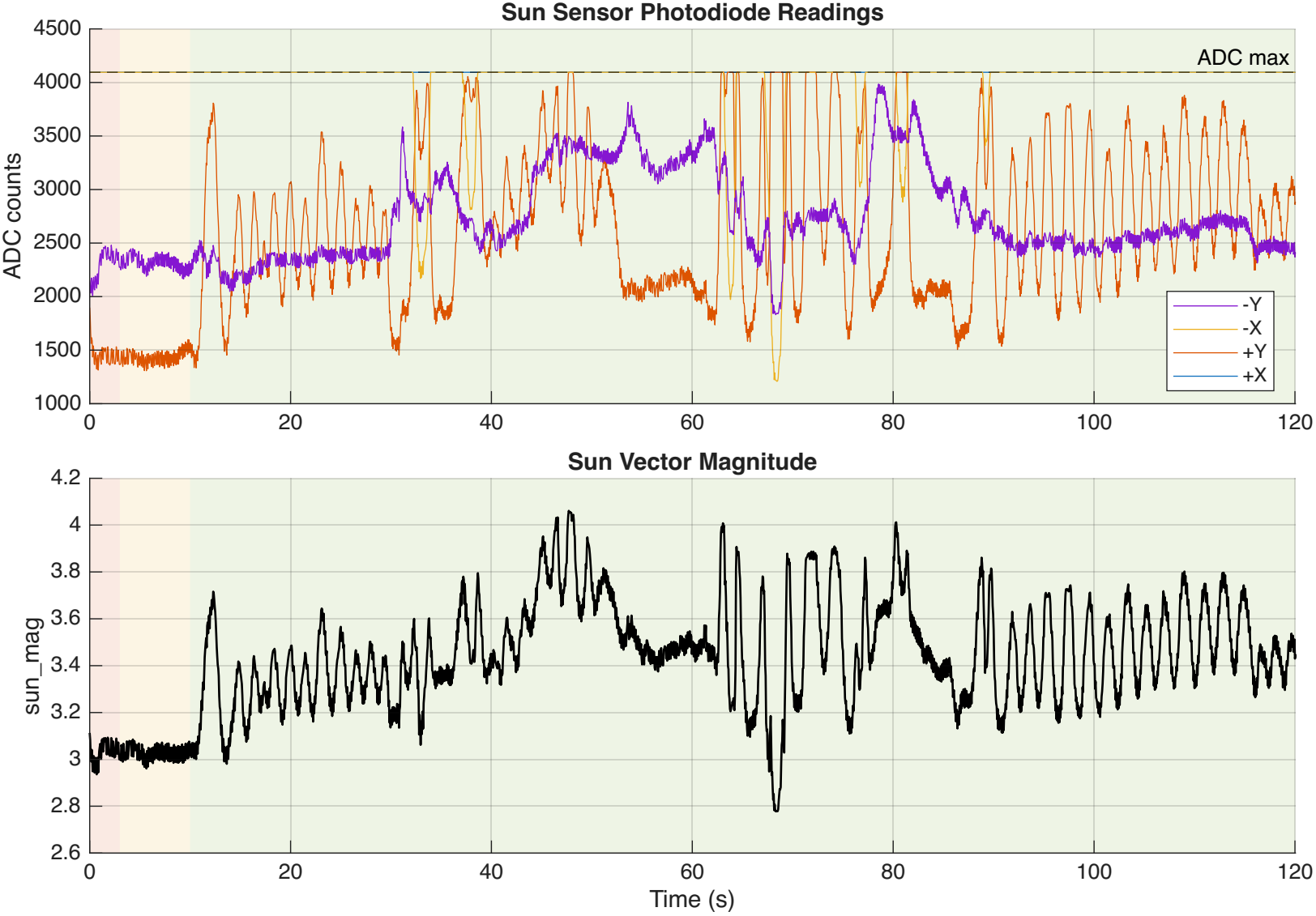
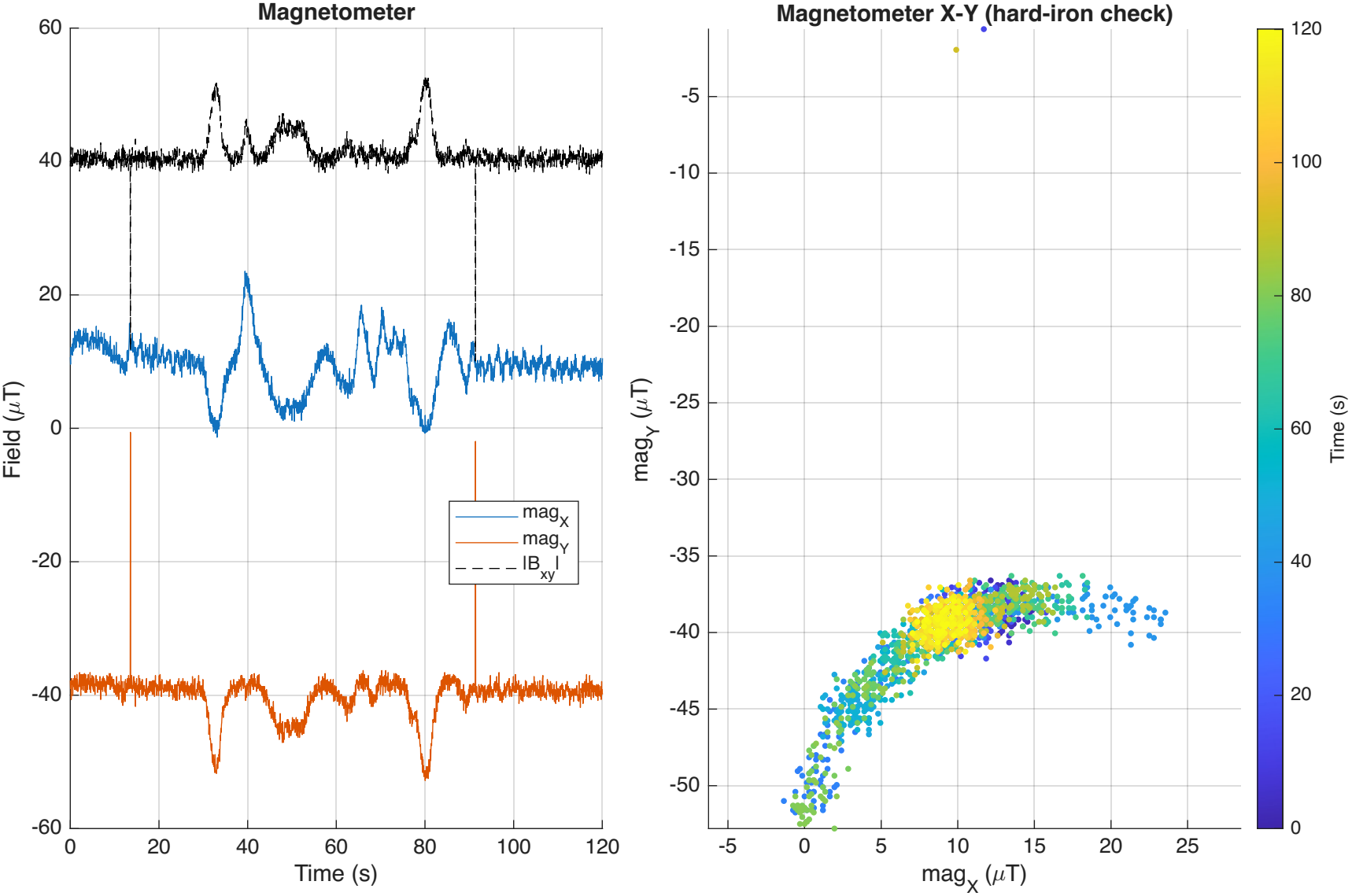
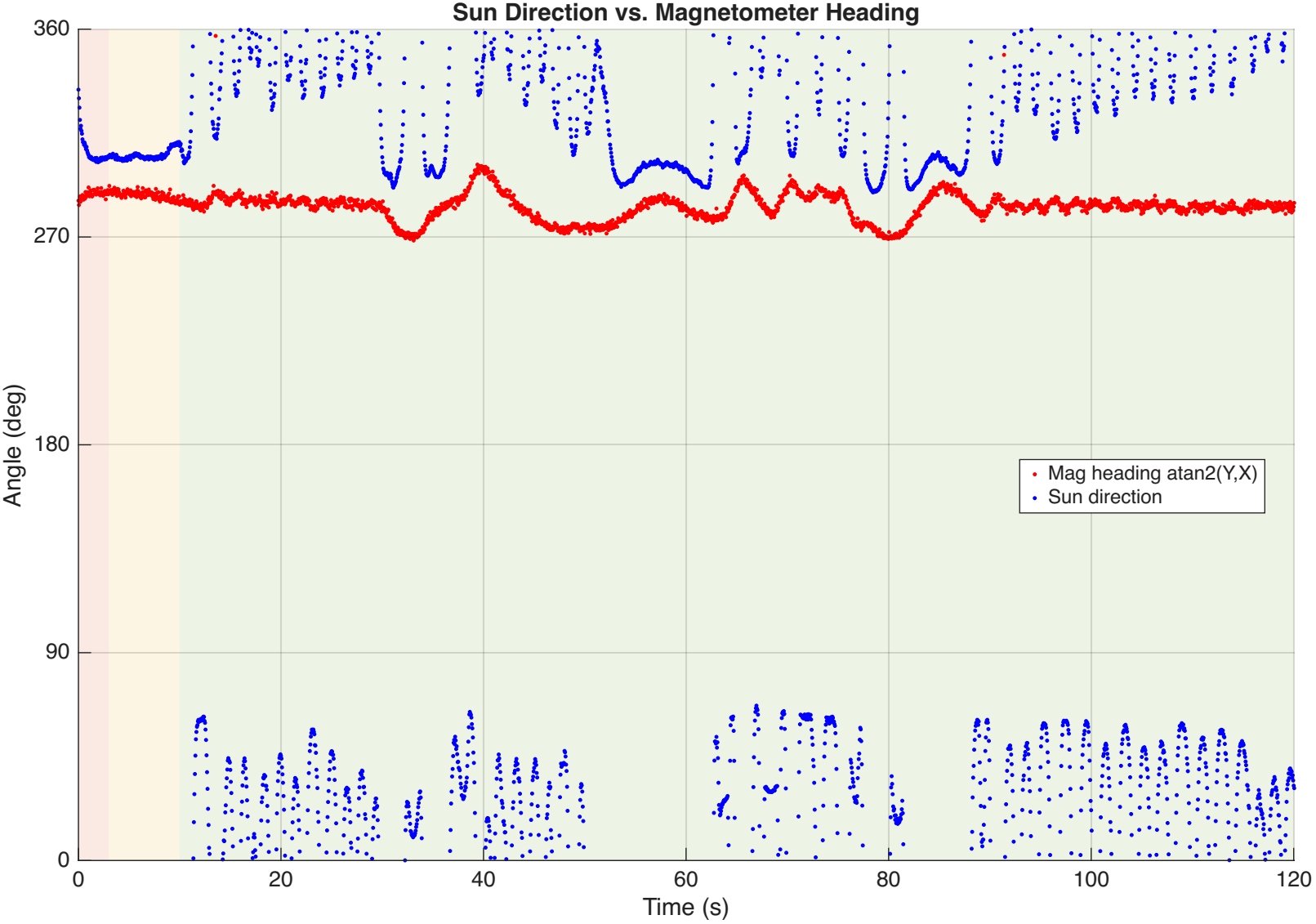

Sun Direction in Body Frame
Sun direction (deg)
360
270
180
90
0
0
20
40
60
80
100
120
Pointing Error
Pointing error (deg)
180
135
90
45
0
-45
-90
-135
-180
SPINUP
HOLD
TRACK
Time (s)

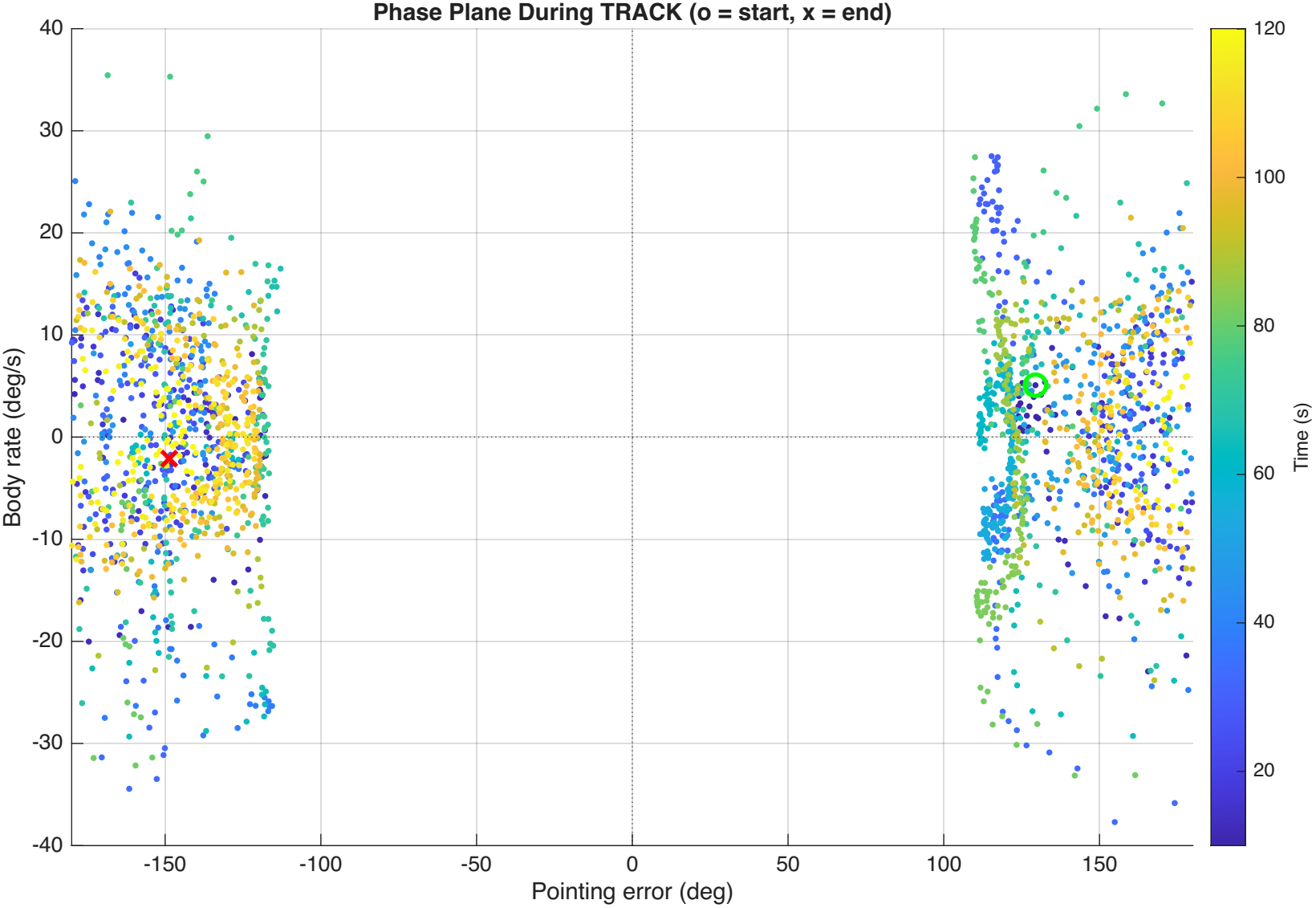

Z-Axis Body Rate (Gyro)
SPINUP
HOLD
TRACK
Body rate (deg/s)
Time (s)
40
30
20
10
0
-10
-20
-30
-40
0
20
40
60
80
100
120

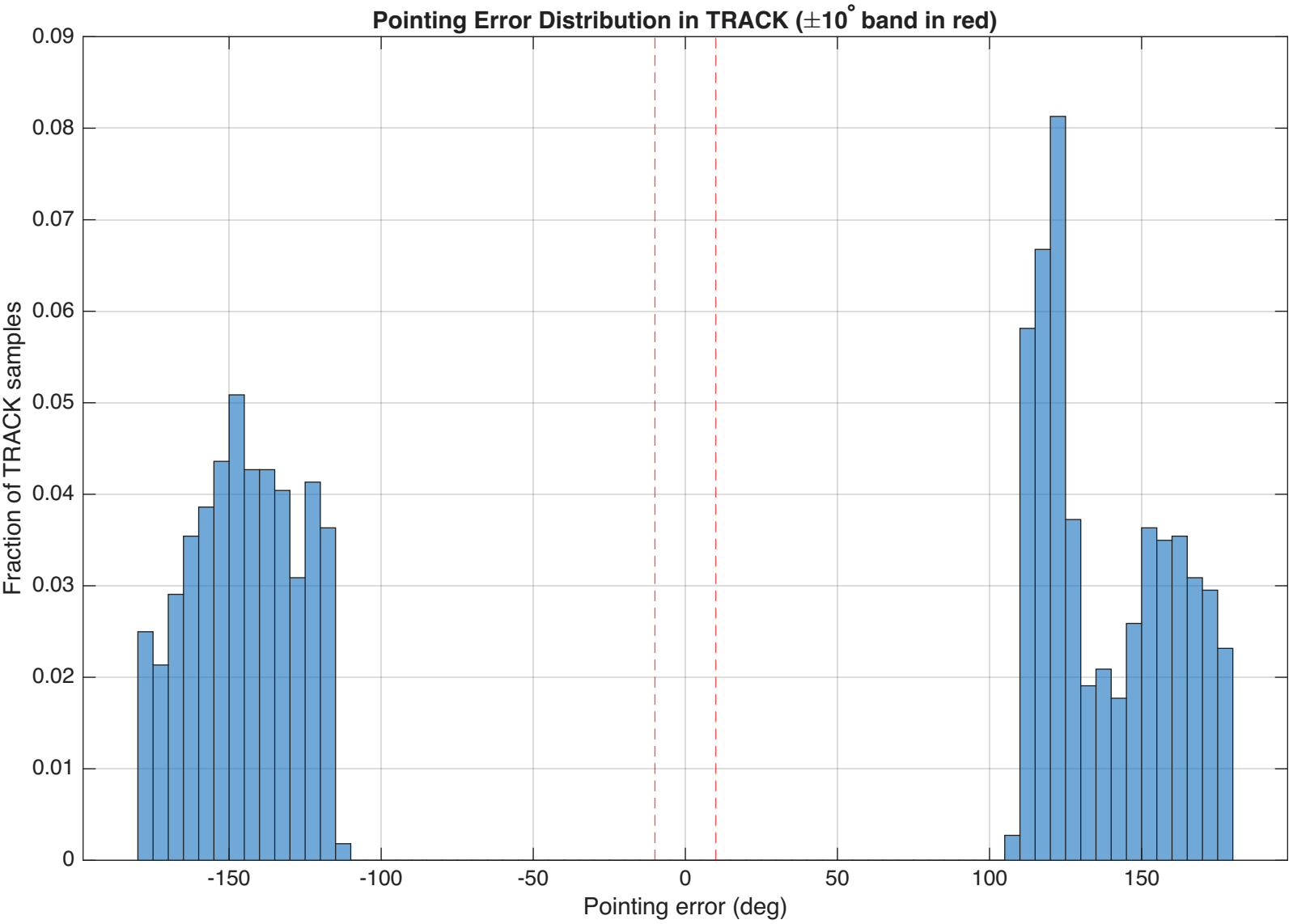

Reaction Wheel Speed
Wheel speed (RPM)
400
300
200
100
0
-100
-200
-300
-400
0
20
40
60
80
100
120
Measured
Commanded
Commanded Torque (magenta = wheel saturated)
Torque cmd (mN·m)
300
200
100
0
-100
-200
-300
Saturated
Torque cmd
Time (s)


Sun Sensor Photodiode Readings
ADC max
ADC counts
4500
4000
3500
3000
2500
2000
1500
1000
0
20
40
60
80
100
120
-Y
-X
+Y
+X
Sun Vector Magnitude
sun_mag
4.2
4
3.8
3.6
3.4
3.2
3
2.8
2.6
Time (s)


Magnetometer
Field (μT)
Time (s)
magX
magY
|Bxy|
Magnetometer X-Y (hard-iron check)
magY (μT)
magX (μT)
Time (s)


Sun Direction vs. Magnetometer Heading
360
270
180
90
0
Angle (deg)
0
20
40
60
80
100
120
Time (s)
Mag heading atan2(Y,X)
Sun direction


Phase Plane During TRACK (o = start, x = end)
Body rate (deg/s)
Pointing error (deg)
Time (s)
40
30
20
10
0
-10
-20
-30
-40
-150
-100
-50
0
50
100
150
120
100
80
60
40
20


Pointing Error Distribution in TRACK (±10° band in red)
Fraction of TRACK samples
0.09
0.08
0.07
0.06
0.05
0.04
0.03
0.02
0.01
0
-150
-100
-50
0
50
100
150
Pointing error (deg)